Powerplant and upgrades study

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1 Purpose and scope

This study answers the question that P00 assigns to A05: what must we not prevent today so that we can grow tomorrow. Its output is the list of provisions for P01 in Section 9 (CRIT-07) and the tow data for REQ-14. It does not choose an engine or a configuration. Two architectures are distinguished, defined by what they represent:

- **A-70:** ALIMOCHE with an engine, within the same legal category as the light glider. Its limit is legal before it is a mass limit: less than 70 kg at take-off without pilot or parachute, with engine, propeller and energy included.
- **A-120:** similar structure, reinforced, with an engine, assessed in another legal category (up to 120 kg). It is another aircraft, not an upgrade of the basic ALIMOCHE, and is included only as a reference.

A05 rule: no provision may compromise the 70 kg of the basic glider (REQ-01).

2 Input data

Table 1 lists the inputs of the study and their origin.

Table 1. Inputs of the study and their origin. Positions measured from the root leading edge, positive aft.

Input	Value	Source	Class
Glider mass without pilot or parachute	< 70 kg	ALM-00 Rev. 0.3; RD 765/2022 art. 2.2 d)	Requirement (REQ-01)
Calculation pilot	90 kg (50, 70 and 90 kg for CG)	ALM-00 Rev. 0.3	Requirement
Wing polar	S = 10 m², AR = 10, CD0 = 0.055 (+0.005 for installation), e = 0.70, CLmax = 1.5	ALM-P00-A02-01 Rev. 0.1	A02 assumption
Calculation condition	Lillo at 40 °C, ρ = 1.04 kg/m³	ALM-P00, ASM-01	Assumption
Reference speed	1.30 Vs (Vs = 51 km/h at 160 kg)	This study	Calculation criterion
Common climb criterion	Vz ≥ 1.0 m/s at the actual mass of each system (sensitivity at 0.7 m/s)	This study	Calculation criterion
Battery	165 Wh/kg pack, up to 10C	Molicel P45B cell: 4.5 Ah, 45 A, 68–70 g; pack factor 0.7	Assumption
Efficiencies	Propeller 0.70 motor and controller 0.90	Typical preliminary-design values	Assumption

Turbines	90 % of catalogue thrust; catalogue consumption at full power	Xicoy X240 and X380 data sheets	Data and assumption
Geometry for CG	Datum at the root leading edge; chord 1.60 m in-flight CG 25–30 % (400–480 mm); pilot at −251 mm	SG-38 flight manual (BWLV)	Similar-aircraft data
ALIMOCHE airframe for CG	62 kg, empty CG at 1221 mm (balances the 70 kg pilot at 27.5 %)	This study	Assumption to be confirmed in P01

3 Legal consequence of adding an engine

Reading of RD 765/2022, consolidated after RD 182/2026. The conclusion belongs to A01, which must check it before 30/10/2026. Table 2 gives the category of each architecture.

Table 2. Legal category of each architecture according to the reading of RD 765/2022. Pending confirmation by A01.

Architecture	Category	How mass counts	Requirements
Basic glider	Glider: not a ULM (art. 2.2 a); EU and national glider regulations (art. 2.3)	Empty mass < 70 kg (REQ-01)	Those set by A01 for gliders
A-70	Aircraft with total take-off weight < 70 kg without pilot or parachute: not a ULM (art. 2.2 d)	Engine, propeller, battery, fuel and ballast count within the 70 kg	Fourth additional provision: harness, helmet in an open aircraft, ≤ 3000 m, outside controlled airspace and urban areas, daytime VFR. Registration, licence and insurance: A01
A-120	Powered aircraft of light structure, ≤ 120 kg at take-off without pilot or parachute: outside the RD (art. 2.1 d)	As A-70, with 120 kg	The same limitations of the fourth additional provision. Registration, licence and insurance: A01
Over 120 kg	Category A ULM (art. 1.2)	—	ULM licence, airworthiness and registration. Outside the scope of ALIMOCHE

Source: RD 765/2022, consolidated text at 12/03/2026.

Adding an engine changes the category in every case: the glider ceases to be one and becomes subject to the limitations of the fourth additional provision. Between A-70 and A-120 the difference is in permissible mass, not in requirements. A01 must also confirm that fuel and battery count (the text speaks of «total take-off weight») and that a tube-and-fabric aircraft fits «light structure».

Preliminary answers for A01. The parachute does not count towards the 70 and 120 kg limits: both articles refer to the mass «less pilot and parachute», and do not distinguish between a personal parachute and an aircraft rescue system, so the latter is understood to be excluded as well. Ballast, on the other hand, does count in A-70. As a loads standard, CS-22 in its current wording raises the emergency-landing loads to 15 g forward, 7.5 g upward, 6 g sideways and 9 g downward (previously 9 / 4.5 / 3 / 4.5 g), and its AMC 22.561 links them to parachute rescue systems. These are the values proposed for the engine mount and the parachute anchorage.

4 Powerplant options and maximum airframe mass

There are three options: none, sustainer (holding height after an external launch) and self-launch (climbing under its own power). In A-70 the decisive question is not which engine, but how much airframe is left after installing it. Figure 1 summarises it with the common criterion: climb of at least 1.0 m/s, each system calculated at its actual mass. Table 3 gives the values for each system.

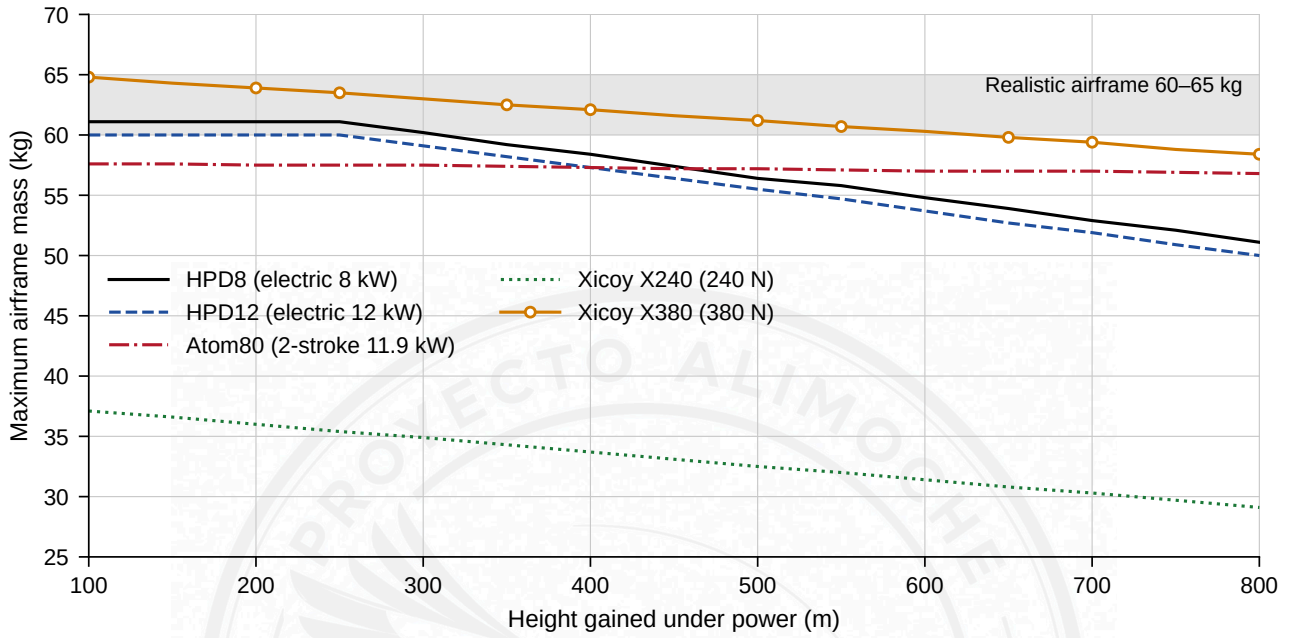


Figure 1. Maximum airframe mass that keeps the powered version below 70 kg, against height gained under power. Without installation reinforcements or landing gear (see Section 5); the grey band is the airframe that A02 considers realistic. Curves digitised from the Rev. 0.5 chart (data file not available); values at 200, 400 and 600 m are those of Table 10.

Source: Appendix A; data/max_airframe_A70.csv.

Table 3. Mass, maximum airframe and cost of each system with the common climb criterion. Cost of engine, controller or ECU, battery or tank, propeller and installation; only the turbines have a complete public price list.

System	Added mass, 60 kg airframe, 400 m (kg)	Max. A-70 airframe at 200 / 400 / 600 m (kg)	Added mass A-120, 100 kg airframe, 400 m (kg)	Installed cost (excl. VAT)
HPD8 (electric 8 kW)	11.7	61.1 / 58.4 / 54.8	Does not climb at 1 m/s	3800–5300 €
HPD12 (electric 12 kW)	12.9	60.0 / 57.3 / 53.7	15.6 (21.2 at 600 m: exceeds 120 kg)	4000–5600 €
Atom80 (2-stroke, 11.9 kW)	12.7	57.5 / 57.3 / 57.0	12.9	2800–3800 €
Xicoy X240 (240 N)	Does not climb at 1 m/s (0.65 m/s)	Rejected	Rejected (0.13 m/s)	3450–3650 €
Xicoy X380 (380 N)	7.8	63.9 / 62.1 / 60.3	9.7	5100–5300 €

Source: Appendix A (Table 10 and Table 11).

Conclusions for P01:

- **A-70 is almost unfeasible:** with a realistic airframe of 60–65 kg, almost no system fits in A-70. Only the X380 allows it: with the reinforcements of Section 5, an airframe of up to 61 kg for 200 m and 59.5 kg for 400 m as a sustainer; with self-launch landing gear, 59 kg and 57.5 kg. Electric and 2-stroke systems require airframes of 53–58 kg.
- **The X240 does not meet the common criterion:** even with a 60 kg airframe it does not exceed 0.65 m/s. Relaxing the criterion to 0.7 m/s does not rescue it either: it would require an airframe of 50 kg or less. At 0.7 m/s the electric systems get worse (the climb lasts longer) and the Atom80 and X380 hardly change.

- **Sustainer:** holding 10 minutes of level flight at 170 kg needs ≈ 8 kg of battery, ≈ 0.6 kg of petrol or 5–8 kg of kerosene. The turbine, light for a short climb, stops being light as soon as the engine runs for more than a few minutes.
- **A-120:** all propeller systems fit within 120 kg except the HPD8 (insufficient power) and the HPD12 at 600 m. It is the architecture in which a powerplant makes sense.
- **Cost:** any installed system costs between 80 % and 160 % of the glider's materials budget (3500 ± 500 €). The 2-stroke is the cheapest; the electric costs about 30 % more than an equivalent 2-stroke, according to the manufacturer itself, mainly because of the battery.

Consequently, the powerplant must not condition P01 beyond the provisions of Section 9. A realistic powered ALIMOCHE is A-120, that is, another aircraft.

5 Installation reinforcements

The fixed mass of the systems (Appendix A) includes only 1.5–1.8 kg of mounting. Installing an engine also requires a mount, transmitting the thrust and crash loads, protecting the pilot and, for self-launch, landing gear. Table 4 estimates these items.

Table 4. Estimated reinforcement mass for installing an engine. Does not include the oversizing of the primary structure for A-120.

Item	Mass (kg)	Applies to
Pylon or mount: two aluminium tubes of ≈ 0.8 m, plate and rubber mounts	1.0–1.8	Sustainer and self-launch
Compression strut to the keel for forward crash loads	0.3–0.6	Sustainer and self-launch
Local reinforcement of the central node and the keel	0.3–0.6	Sustainer and self-launch
Firewall (2-stroke) or fireproof battery box (electric)	0.2–0.5	Sustainer and self-launch
Landing gear: 200–260 mm wheel, axle, fork and skid reinforcement	1.5–2.5	Self-launch only
Total	1.8–3.5 (sustainer) · 3.3–6.0 (self-launch)	

Source: author's estimate.

The load cases that size these parts are three. The crash: with 15 g forward, a 12.7 kg 2-stroke behind the pilot generates ≈ 1.9 kN ultimate (1.1 kN with the former 9 g), a 5.5 kg electric ≈ 0.8 kN and the X380 ≈ 0.6 kN. The engine thrust and torque, plus the propeller gyroscopic loads, which only affect propeller systems. And landing with more mass, which in self-launch falls on the landing gear. Subtracting the central values (2.65 kg as a sustainer and 4.65 kg with landing gear), the maximum A-70 airframe is as shown in Table 5.

Table 5. Maximum airframe compatible with A-70 including installation reinforcements. The X240 is excluded in every case.

System	200 m: no reinforcement / sustainer / with gear (kg)	400 m: no reinforcement / sustainer / with gear (kg)
HPD8 (electric 8 kW)	61.1 / 58.4 / 56.4	58.4 / 55.7 / 53.7
HPD12 (electric 12 kW)	60.0 / 57.3 / 55.3	57.3 / 54.6 / 52.6
Atom80 (2-stroke 11.9 kW)	57.5 / 54.9 / 52.9	57.3 / 54.6 / 52.6
Xicoy X380 (380 N)	63.9 / 61.3 / 59.3	62.1 / 59.5 / 57.5

Source: refuerzos.py; datos/celula_max_con_F4.csv.

With reinforcements included, no propeller system leaves more than 58.4 kg of airframe, and the X380 only exceeds 60 kg as a sustainer at 200 m. The input for P01 is clear: a powered A-70 requires an airframe of 53–61 kg depending on system and mission, below what A02 considers realistic.

6 CG position and volume

The geometry and limits of the SG-38 flight manual are used: datum at the root leading edge, chord 1.60 m and in-flight CG between 25 % and 30 % (400–480 mm). The pilot arm, −251 mm, is derived from the manual itself: a 110 kg airframe with the empty CG at 880 mm and a 70 kg pilot give the normal CG of 440 mm. For ALIMOCHE a 62 kg airframe balanced for the same 70 kg pilot is assumed. Ballast is placed, as on the SG-38, forward under the pedals (−1200 mm) or aft on the rear strut (+1500 mm). Table 6 gives the result for each installation.

Table 6. CG position, ballast required and volume of each installation. Permissible SG-38 range 25–30 %; «fwd» forward ballast, «aft» rear ballast; ballast counts within the 70 kg in A-70.

Installation	Added mass (kg)	CG with 90 / 70 / 50 kg pilot (% chord)	Ballast with 90 kg / 50 kg pilot (kg)	Volume and fit in an SG-38-type airframe
Basic glider (no engine)	0.0	21.8 / 27.5 / 35.2	7.0 aft / 5.6 fwd	With 50–90 kg of pilot the CG moves from 22 % to 35 %: more than the SG-38 range
A Electric: motor and propeller on pylon + battery under the seat	12.9	23.2 / 28.5 / 35.6	4.4 aft / 6.7 fwd	Battery ≈ 3–4 L under the seat; propeller Ø 1.0–1.2 m. Fits with pylon
B Electric: motor, propeller and battery on pylon	12.9	26.7 / 32.5 / 40.2	0 / 12.2 fwd	Bulky pylon, far aft
C 2-stroke on pylon + tank behind the seat	12.7	27.1 / 33.0 / 40.9	0 / 12.9 fwd	Engine ≈ 40 × 35 × 30 cm on pylon. Far aft
D 2-stroke tractor in the nose	12.7	10.9 / 14.6 / 19.5	33.8 aft / 10.1 aft	Propeller in front of the pilot's feet; CG far forward. Rejected
E 2-stroke pusher behind the seat (under the wing)	12.7	22.9 / 28.3 / 35.3	4.9 aft / 6.4 fwd	The propeller disc crosses the keel: forces a tail redesign
F X380 turbine on pylon + tank behind the seat	7.8	22.7 / 28.2 / 35.5	5.4 aft / 6.3 fwd	Turbine Ø 13 cm; tank ≈ 5 L behind the seat. Jet over the tail

Source: SG-38 flight manual (BWLV); centrado.py; datos/centrado.csv.

The most important result does not depend on the engine: with a 62 kg airframe the pilot weighs more than the aircraft, and the CG moves from 22 % to 35 % of the chord between 50 and 90 kg pilots. With the SG-38 range up to 7 kg of ballast would be needed, just as the SG-38 needs up to four weights of 2.6 kg. The way out is to widen the CG range to ≈ 22–35 % (tail and stability sized for it) or to provide forward and rear ballast housings. Moving the seat is not enough: about 27 cm of travel would be needed.

As for the engine, installations A, E and F (heavy masses next to the pilot and light ones on the pylon) hardly change the balance of the basic glider. B and C, with all the mass on the pylon, require 12–13 kg of forward ballast with a 50 kg pilot, which in A-70 is unacceptable. The engine in the nose is rejected.

7 Safety by family

Table 7 lists the main risks of each engine family and what they imply for the design.

Table 7. Main risks by engine family.

Family	Main risk	What it implies for the design
All	Engine failure in a climb at low height	Minimum decision height and a field within reach (A04). The aircraft must glide with the engine stopped without abrupt trim changes

Propeller (electric or 2-stroke)	Propeller close to the pilot's head, to cables and to the tail	Clear disc with ≥ 15 cm to structure and cables; guard if within the pilot's reach
2-stroke	Fuel and fire next to the pilot in an open airframe; vibration and fatigue	Protected tank behind the seat, firewall, rubber mounts, no fabric or Dyneema near the exhaust
Electric	Battery fire; high-power discharge	Fireproof, ventilated box, BMS, disconnector within the pilot's reach
Turbine	Gases at several hundred degrees; rotor above 100 000 rpm without containment; slow response	Jet clear of fabric, cables and tail; shield in the rotor plane; not to be used as a take-off engine
All, in a crash	Engine behind the pilot in a frontal impact	Mount and strut for 15 g forward (current CS-22); engine mass as low and as close to the CG as possible

Source: own work; X240: 113 000 rpm according to the manufacturer's data sheet; CS-22, CS 22.561.

8 Parachute, thermals and tow

Parachute. There are three options. The pilot's personal parachute (≈ 6–7 kg) requires leaving the aircraft and is of no use at bungee-launch heights. The rocket-deployed ballistic system is rejected for mass and cost. The recommended one is a hang-glider rescue system for the whole aircraft, anchored to the central node: for example, the Charly DIAMONDcross delta light 220 accepts 220 kg with a mass of 2.46 kg, with margin against the 160–170 kg of ALIMOCHE. It is only useful above ≈ 100 m, that is, with a tow launch. The provision is an anchorage fitting at the central node sized for the opening load given by the manufacturer and for the cases of CS 22.561, and a pilot exit free of cables. According to the reading of the legal section, it does not count within the 70 kg.

Thermals. With the A02 polar the minimum sink rate is 1.50 m/s, at the stall, and 1.71 m/s at 1.2 Vs, which is the realistic speed for turning. A normal thermal requires ≲ 1.0 m/s. Lowering CD0 to 0.040 with a fairing gives 1.46 m/s extending the span to 12 m, 1.43 m/s both, 1.21 m/s. ALIMOCHE will not be a thermal aircraft; with improvements it could use ridge lift and strong thermals. The doors not to close are attachment points for a fairing and a wing that accepts tip extensions. A pylon and a folded propeller add ≈ +0.05 m/s.

Tow (REQ-14). With Vs ≈ 51–52 km/h, the tow would be flown at 1.3–1.5 Vs, that is, 66–78 km/h: the maximum tow speed set by P01 must cover at least 80 km/h. For reference, the SG-38 (Vs = 40 km/h) allows 60 km/h on winch or car and 90 km/h on aerotow, with normal towing at 60–90 km/h. Car or winch tow is compatible; it needs a CG hook with safety release and a weak link. The SG-38 has, as this study proposes, a nose hook for bungees and a safety release for winch and aerotow. Its weak links are 300 daN for aerotow and 400 daN for winch and car at 210 kg scaled to 160–170 kg they give ≈ 2.4 kN and ≈ 3.2 kN, above the rule of thumb of 1.3 times the weight (≈ 2 kN).

Aerotow needs a slow tug. A hang-glider tug such as the Bailey-Moyes Dragonfly, designed to tow hang gliders and ultralight gliders, stalls at 31 km/h, climbs at 51 km/h and cruises at 74 km/h, a band that covers the tow speeds of ALIMOCHE. The conventional tugs of aero clubs fly the tow faster; A04 must confirm which are available and at what speeds.

Take-off and ground roll. A coefficient μ ≈ 0.05 is only valid for a wheel on a hard runway. Table 12 sweeps μ: with a wheel on grass all propeller systems take off in 100–150 m, and the X380 from 150 m; with a skid on grass (μ ≈ 0.3) only the HPD12 and the Atom80, and with a 150 m run. For self-launch, therefore, a wheel is needed, and the turbine also needs a hard runway or a long run.

9 List for P01 (CRIT-07)

Table 8 lists the provisions that P01 must not prevent. All of them fit in the basic glider without compromising the 70 kg.

Table 8. Provisions that P01 must not prevent. Estimated masses; total ≈ 0.5–0.8 kg without ballast. Does not include the reinforcement of the primary structure for A-120, which belongs to another aircraft.

#	Provision	Purpose	Architecture	Mass (g)
1	Hard point with a load path at the central node for a pylon or mount, with holes or plate provided and an attachment point for a strut to the keel (15 g forward)	Engine on pylon	A-70 and A-120	100–150
2	Free volume of ≈ 5 L under or behind the seat, near the CG, with four attachment points	Battery, tank or ballast	Both	≈ 50
3	CG range of ≈ 22–35 % of chord verified with 50 and 90 kg pilots, or forward and rear ballast housings of up to ≈ 7 kg	50–90 kg pilots and engine on pylon	Both	50–100
4	Disc of Ø 1.2 m behind the trailing edge, above the keel, clear of cables and tubes, with ≥ 15 cm to tail, ground and pilot's head	Pusher propeller	Both	0 (geometry)
5	Cable runs for throttle, engine cut-off and power	Controls and wiring	Both	≈ 20
6	Fittings for nose and CG hooks, with a load path for weak links of ≈ 2.4 kN (aerotow) and ≈ 3.2 kN (car and winch)	Tow (REQ-14)	Both	150–250
7	Skid with provision for a wheel compatible with self-launch	Self-launch	Both	≈ 50
8	Parachute anchorage fitting at the central node and pilot exit free of cables	Parachute	Both	50–100
9	Attachment points for a fairing or nacelle and a wing that accepts tip extensions	Thermals	Glider	≈ 50

10 Open items

Table 9 lists the pending data, who is responsible and the due date.

Table 9. Pending data, responsibilities and dates.

Data	Purpose	Responsible	Date
Confirm the reading of RD 765/2022 (Table 2) and the preliminary answers of the legal section; registration, licence and insurance	Legal section and parachute	A01 · [name]	30/10/2026
Tow load cases and strength of the hook fitting (CS 22.581, 22.583 and 22.585)	Provision 6	A01 · [name]	30/10/2026
Polar extrapolated to 210 kg (A-120) and landing speed	Appendix A	A02 and A04 · [name]	30/10/2026
Available tugs and their speeds; μ measured at Lillo with skid and with wheel	Tow; Table 12	A04 · [name]	30/10/2026
Geiger quote (no public price list), dated prices excl. VAT and filed data sheets	Table 3 and Table 11	A03 · [name]	30/10/2026
Airframe geometry, CG range, ballast system and installation reinforcements	Section 5 and Section 6	P01	P01

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Appendix A Supporting calculations

Screening model, reproducible with modelo.py, refuerzos.py, centrado.py and rodadura.py. The drag is given by Eq. 1 at the reference speed 1.30 Vs:

$$D = \frac{1}{2}\rho V^2 S(C_{D0} + \Delta C_{D0}) + \frac{2k(mg)^2}{\rho V^2 S}, \quad k = \frac{1}{\pi \ \mathrm{AR} \ e} \tag{1}$$

For a propeller, $P_{\mathrm{shaft}} = (DV + mgV_z)/\eta_p$ with $V_z = 1.0$ m/s. For a turbine, $V_z = (T - D)V/(mg)$ at the best speed between 1.30 V_s and 100 km/h, with 90 % of catalogue thrust; if it does not reach 1.0 m/s, the system is rejected. Each system is calculated at its actual mass (airframe + system + energy + pilot), closing the mass by iteration.

The battery meets energy and power at the same time, Eq. 2, with 165 Wh/kg and 10C, 80 % usable energy and 15 % reserve. The 2-stroke uses 0.60 kg/kWh; the turbines, their catalogue consumption at full power (X240 0.600 kg/min X380 0.985 kg/min), with the same reserve. The screening results are given in Table 10.

$$m_b \geq \max\left(E/e_{\mathrm{pack}}, P_{\mathrm{elec}}/\left(C \cdot e_{\mathrm{pack}}\right)\right) \tag{2}$$

where m_b is the battery mass, E the energy required, e_{pack} the pack specific energy, P_{elec} the electrical power and C the discharge rate.

Table 10. Screening results with the common climb criterion. The maximum airframe of the X240 (31–37 kg) is not realistic. With a 100 kg airframe (A-120) the A02 polar is extrapolated to ≈ 210 kg (21 kg/m², Vs ≈ 58 km/h), pending confirmation by A02.

System	Height (m)	Added mass, 60 kg airframe (kg)	Max. A-70 airframe (kg)	Added mass, 100 kg airframe (kg)	Vz with 60 kg airframe (m/s)
HPD8 (electric 8 kW)	200	8.9	61.1	No: power	1.00
HPD8 (electric 8 kW)	400	11.7	58.4	No: power	1.00
HPD8 (electric 8 kW)	600	15.7	54.8	No: power	1.00
HPD12 (electric 12 kW)	200	10.0	60.0	11.7	1.00
HPD12 (electric 12 kW)	400	12.9	57.3	15.6	1.00
HPD12 (electric 12 kW)	600	17.0	53.7	21.2	1.00
Atom80 (2-stroke 11.9 kW)	200	12.5	57.5	12.6	1.00
Atom80 (2-stroke 11.9 kW)	400	12.7	57.3	12.9	1.00
Atom80 (2-stroke 11.9 kW)	600	13.0	57.0	13.3	1.00
Xicoy X240 (240 N)	200	No: Vz 0.65 m/s	36.0	No: Vz 0.13 m/s	0.65
Xicoy X240 (240 N)	400	No: Vz 0.65 m/s	33.7	No: Vz 0.13 m/s	0.65
Xicoy X240 (240 N)	600	No: Vz 0.65 m/s	31.4	No: Vz 0.13 m/s	0.65
Xicoy X380 (380 N)	200	6.0	63.9	6.9	2.17
Xicoy X380 (380 N)	400	7.8	62.1	9.7	2.13
Xicoy X380 (380 N)	600	9.7	60.3	12.8	2.09

Source: modelo.py; datos/masas_cribado.csv.

Sensitivity to the climb criterion. With Vz ≥ 0.7 m/s, the maximum airframe at 400 m becomes 56.3 kg (HPD8), 55.2 kg (HPD12), 57.1 kg (Atom80), 62.1 kg (X380) and 50.1 kg (X240). The slower climb demands more energy from the electric systems and does not change the order or the conclusion. Table 11 breaks down the installed cost.

Table 11. Breakdown of installed cost. Propeller, installation and electric controller are estimates; the HPD8 is estimated at 3800–5300 € and the X240 at 3450–3650 € (3249 € list price).

Item (excl. VAT)	HPD12 electric	Atom80 2-stroke	Xicoy X380
Motor and controller or ECU	3000–4000 € (no public price list; +30 % over an equivalent 2-stroke according to Geiger)	2300–2900 € (2700–3040 USD at a US distributor)	4899 € (Xicoy price list)
Energy	1.2 kWh battery: 76 P45B cells at 3.76 € ≈ 290 €, plus BMS, box and connectors: 550–750 €	Tank: 50–100 €	Tank and header tank: 50–100 €
Propeller	Carbon, Ø 1.2 m: 300–500 €	Carbon, Ø 1.25 m: 300–500 €	—
Installation and reinforcements	150–300 €	150–300 €	150–300 €
Total	4000–5600 €	2800–3800 €	5100–5300 €

Source: Xicoy (25/10/2026); aviatorppg.com, nkon.nl and geigerengineering.de (26/10/2026).

Table 12 gives the mean thrust needed for self-launch against the rolling surface.

Table 12. Mean thrust needed for self-launch against ground roll. 170 kg, take-off at 1.1 Vs (58 km/h), mean lift during the run 30 % of the weight. Available thrust: X240 216 N and X380 342 N (90 % of catalogue); propellers estimated by momentum theory, HPD8 ≈ 420 N, HPD12 and Atom80 ≈ 550 N, decreasing with speed. Skid-on-grass μ estimated.

Surface	μ (–)	Run (m)	Mean thrust needed (N)	Systems that achieve it
Hard runway, wheel	0.04	100	307	HPD8, HPD12, Atom80, X380 (90 %)
Hard runway, wheel	0.04	150	234	HPD8, HPD12, Atom80, X380 (90 %)
Short grass, wheel	0.08	100	354	HPD8, HPD12, Atom80
Short grass, wheel	0.08	150	280	HPD8, HPD12, Atom80, X380 (90 %)
Grass, skid	0.30	100	610	None
Grass, skid	0.30	150	537	HPD12, Atom80

Source: rodadura.py; datos/rodadura.csv; energy bound of the take-off run.